EXPERIMENT-13

Q1.Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:
Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2
Find the sum of its even digits and the sum of its odd digits separately. E.g. Number: 12345, Sum_even = 6, Sum_odd = 9

1. Algorithm:

Algorithm

1. Start.
2. Input a positive integer "num".
3. Display the menu:
   - 1. Count even and odd digits.
   - 2. Find the sum of even and odd digits.
4. Input the user's choice.
5. Use a "switch" statement to perform the selected operation.

Case 1: Count Even and Odd Digits

6. Initialize "even_count = 0" and "odd_count = 0".
7. Copy "num" into a temporary variable "temp".
8. Repeat while "temp > 0":
   - Find the last digit using "digit = temp % 10".
   - If "digit % 2 == 0", increase "even_count" by 1.
   - Otherwise, increase "odd_count" by 1.
   - Remove the last digit using "temp = temp / 10".
9. Display "even_count" and "odd_count".

Case 2: Sum of Even and Odd Digits

10. Initialize "sum_even = 0" and "sum_odd = 0".
11. Copy "num" into "temp".
12. Repeat while "temp > 0":

- Find the last digit using "digit = temp % 10".
- If "digit % 2 == 0", add "digit" to "sum_even".
- Otherwise, add "digit" to "sum_odd".
- Remove the last digit using "temp = temp / 10".

13. Display "sum_even" and "sum_odd".
14. If the choice is neither 1 nor 2, display ""Invalid choice"".
15. Stop.

2.Test Case Table:

Case	Number	Choice	Expected Output	Actual Output	Result
1	12045	1	Even digits = 3, Odd digits = 2	Even digits = 3, Odd digits = 2	Pass
2	12345	1	Even digits = 2, Odd digits = 3	Even digits = 2, Odd digits = 3	Pass
3	2468	1	Even digits = 4, Odd digits = 0	Even digits = 4, Odd digits = 0	Pass
4	12345	2	Sum(even) = 6, Sum (odd) = 9	Sum (even) = 6, Sum (odd) = 9	Pass
5	2468	2	Sum (even) = 20, Sum(odd) = 0	Sum (even) = 20, Sum (odd) = 0	Pass
6	1357	2	Sum (even)= 0, Sum (odd)= 16	Sum (even) = 0, Sum (odd)= 16	Pass

3. Program, Compilation and Execution:

Session Actions Edit View Help

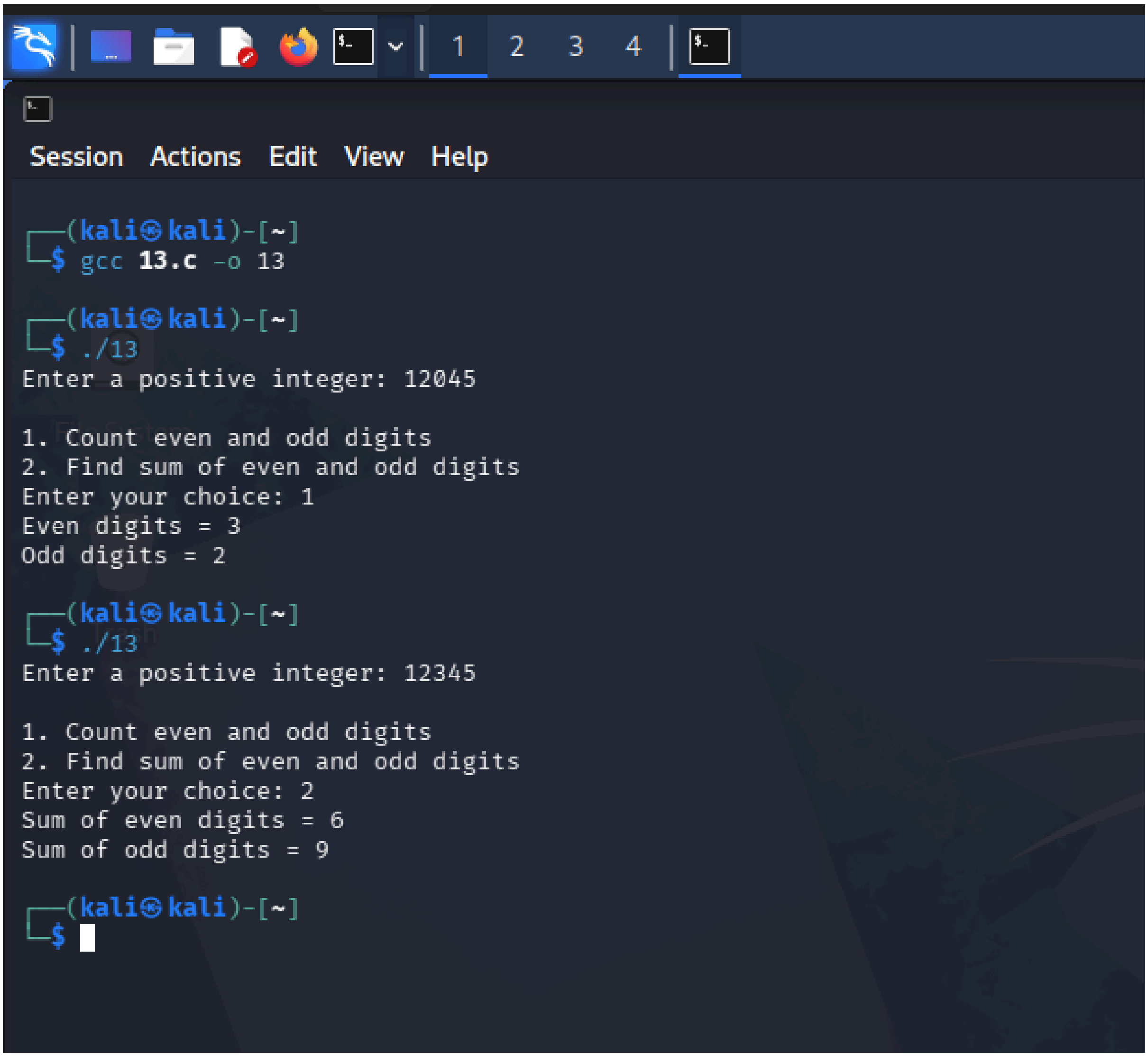
GNU nano 9.0

```c
#include <stdio.h>

int main()
{
    int num, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n1. Count even and odd digits");
    printf("\n2. Find sum of even and odd digits");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    temp = num;

    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;

                temp = temp / 10;
            }

            printf("Even digits = %d\n", even_count);
            printf("Odd digits = %d\n", odd_count);
            break;
```

1 2 3 4

Session Actions Edit View Help

```
  GNU nano 9.0
        case 2:
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;

                temp = temp / 10;
            }

            printf("Sum of even digits = %d\n", sum_even);
            printf("Sum of odd digits = %d\n", sum_odd);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```



4. Result:

The menu-driven C-program was compiled and executed successfully. It correctly uses a switch-case statement to count even and odd digits or calculate their sum separately. All the prepared test cases produced the expected outcomes. One important point: 0 is an even digit, so 2, 0, 4 give an even digit count of 3.

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